

Jessica Robbins

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 <https://jexsaga.github.io/Portfolio/>

About

Energetic and creative Master's student with real-world software development experience and a passion for Animation, VFX, and Game Development. Skilled at learning new tools quickly and bringing technical ideas to life through interactive and visual experiences.

Education

École polytechnique & ENS Louis-Lumière, Paris, France.

September 2026 - Present

- **M2: Master of Science and Technology in Extended Cinematography**
- Expected graduation: August 2027

Institut Polytechnique de Paris, Palaiseau, France.

September 2025 - August 2026

- **M1: Interaction, Graphics and Design; Master's in Computer Science**
- Applicable courses: Interactive 3D Application Development, Fundamentals of Computer Graphics, Computer Animation, Human-Computer Interaction for Mixed Reality, Web Development

College of Science of Engineering, University of Minnesota, Twin Cities USA. GPA: 3.5

September 2021 - May 2025

- **Bachelor of Science in Computer Science** with a minor in French
- Honors: Dean's List, 2022-2023 Academic Year
- Applicable courses: Intro: Artificial Intelligence, Programming Graphics and Games, User Interface Design, Computer Graphics 1

Université Paul Valéry, Montpellier France

January 2023 - May 2023

- Study Abroad - Advanced Track, focused on enhancing language and cultural skills for my French minor

Experience

Game Programming Intern - Cyanide, Nanterre France

April 2026 - Present

- Develop game features in C++ and a proprietary scripting language for Pro Cycling Manager.
- Collaborate closely with senior programmers to design, implement, and maintain game systems.
- Create and integrate user interface features from design mockups using internal development tools.
- Investigate, reproduce, and resolve bugs in coordination with QA and other development teams.
- Participate in feature implementation, testing, and iteration throughout the development cycle.

Software Engineer - Dignified Living Home Care, West St. Paul, MN

May 2025 - December 2025

- Served as the primary developer for internal software used to manage healthcare visit data and operational workflows.
- Designed and developed a web application using .NET, C#, and HTML that enabled staff to track, view, and update visit records in real time.
- Integrated REST APIs and Azure SQL databases to automate data transfer from third-party systems and reduce manual data entry.
- Collaborated directly with administrative and care staff to gather requirements and deliver software solutions tailored to healthcare operations.
- Ensured HIPAA-compliant handling of sensitive patient information while participating in Agile Scrum development and troubleshooting efforts.

Software Engineer - AtriCure, Minnetonka, MN

August 2024 - May 2025

- Continued work commenced in Engineering Co-op.
- Advanced full-stack software solutions for medical devices on Linux, leveraging C, Python, and JavaScript to enhance functionality and reliability.

- Utilized Azure DevOps for version control, code reviews, and collaborative development workflows.
- Improved and maintained the usability of the graphical user interface in Vue with user feedback.
- Proactively implemented enhancements based on user and engineering feedback, improving workflow efficiency and product usability.

Engineering Co-op - AtriCure, Minnetonka, MN

January 2024 - August 2024

- Developed full stack software solutions using C, Python, and JavaScript for medical devices, demonstrating critical thinking in troubleshooting and optimizing code for reliability.
- Designed and implemented two user interfaces in Vue, significantly improving usability and functionality.
- Managed backend processes, including data handling and communication protocols, troubleshooting issues related to GPS, EEPROM emulation, and CAN bus, illustrating swift learning and problem-solving skills.
- Collaborated effectively within cross-functional teams, demonstrating strong teamwork while proactively contributing to projects across various teams.

Skills

Programming Languages & Frameworks: OpenGL, WebGPU, Blender, Unity, Unreal Engine, Python, C, C++, Java, Web Development, Vue, TypeScript, OCaml, JavaScript, MySQL, Azure SQL, .NET, ASP.NET MVC, REST APIs, Agile, SCRUM, Microsoft DevOps, Azure

Spoken Languages: English, French (Upper Intermediate)

Projects:

Video Game - Personal Project

Current

- In collaboration with another student, this game is a Unity project to get to know the tools for game development. This project has commenced with building a robot in Blender and basic game set up.

Character animation - Research Project

Current

- Conducting a solo research project focused on physically-based character animation.

Screen-Space Perceptual Rendering of Human Skin - Image Synthesis Course

March 2026

- Solo project to implement the model used in a research paper in WebGPU.

HCI for Extended Reality Labs and Project - HCI for XR Course

November 2025- January 2026

- Unity projects with a fellow student to develop VR prototypes, learn XR design principles, and iteratively improve usability through testing.

Photo Mosaic Generator - Image Analysis Course

November 2025 - December 2025

- Project with a classmate to develop and test a photo mosaic generator that uses color matching and edge detection, exploring algorithms from research papers and presenting findings.

Computer Animation Labs - Computer Animation Course

September 2025 - December 2025

- OpenGL projects simulating real life phenomena. Some examples include cloth rendering, interactive deformable mesh, water droplet, and collisions.

Ray Tracing - Computer Graphics Course

January 2025 - May 2025

- An academic solo project in C++ to create from scratch an image renderer using raytracing and text file input.

Hole in the Ground Game - Computer Graphics Course

October 2024

- An academic solo project using TypeScript and GopherGfx to create an interactive 3D physics video game.

Dancing Ants - Computer Graphics Course

November 2024

- An academic solo project using GopherGfx to animate ants performing dance moves based on motion capture.

See the linked portfolio for more projects.